

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "Guide to Using Scriptor Modules" document.

Device Specifications

Device Type: Display
Manufacturer: Sony
Firmware Version: N/A
Model(s): FW-98BZ50L, FW-85BZ40L, FW-75BZ40L, FW-65BZ40L, FW-55BZ40L, FW-98BZ30L, FW-85BZ30L, FW-65BZ30L, FW-55BZ30L, FW-50BZ30L, FW-43BZ30L, FW-75BZ30L, FW-85BZ35L, FW-75BZ35L, FW-65BZ35L, FW-55BZ35L

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.17.0000-b002	2.21.0.5

Version History

Module Version	Date	Notes
1_2_0_0	4/25/2024	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: InterfaceName.Unidirectional = 'True'
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: InterfaceName.connectionCounter = 5
- For Serial control, a command pacing of 0.5 seconds is recommended between all commands.

Supported Classes and Examples

SerialClass
InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='FW-98BZ50L')
SerialOverEthernetClass
InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='FW-98BZ50L')
EthernetClass
InterfaceName = ModuleName.EthernetClass('192.168.254.254', 20060, Model='FW-98BZ50L')

Control Commands (Serial)

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command	Value	Value	Value
AspectRatio	'4:3'	'4:3 Wide Zoom'	'16:9'
# AspectRatio example InterfaceName.Set('AspectRatio', '4:3')			
Command	Value	Value	
AudioMute	'On'	'Off'	
# AudioMute example InterfaceName.Set('AudioMute', 'On')			
Command	Value	Value	
AVMute	'On'	'Off'	
# AVMute example InterfaceName.Set('AVMute', 'On')			
Command	Value	Value	Value
Input	'Video / SCART 1'	'HDMI 1'	'HDMI 2'
	'HDMI 3'	'HDMI 4'	'PC'
	'Shared Input'		
# Input example InterfaceName.Set('Input', 'Video / SCART 1')			
Command	Value	Value	Value
MenuNavigation	'Up'	'Down'	'Left'
	'Right'	'Select'	'Return'
	'Home'		
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Up')			
Command	Value	Value	
Power	'On'	'Off'	
# Power example InterfaceName.Set('Power', 'On')			
Command	Value	Value	
VideoMute	'On'	'Off'	
# VideoMute example InterfaceName.Set('VideoMute', 'On')			
Command	Value		
Volume	0 to 100 in steps of 1		
# Volume example InterfaceName.Set('Volume', 100)			

Status Available (Serial)

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},
FeedbackHandler)
```

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command	Value	Value	
AVMute	'On'	'Off'	
# AVMute example InterfaceName.Update('AVMute') Value = InterfaceName.ReadStatus('AVMute') InterfaceName.SubscribeStatus('AVMute', None, FeedbackHandler)			
Command	Value	Value	
ConnectionStatus	'Connected'	'Disconnected'	
# ConnectionStatus example Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command	Value	Value	Value
Input	'Video / SCART 1'	'HDMI 1'	'HDMI 2'
	'HDMI 3'	'HDMI 4'	'PC'
	'Shared Input'		
# Input example InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command	Value	Value	
Power	'On'	'Off'	
# Power example InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)			
Command	Value		
Volume	0 to 100 in steps of 1		
# Volume example InterfaceName.Update('Volume') Value = InterfaceName.ReadStatus('Volume') InterfaceName.SubscribeStatus('Volume', None, FeedbackHandler)			

Control Commands (Ethernet)

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command	Value	Value	
AudioMute	'On'	'Off'	
# AudioMute example InterfaceName.Set('AudioMute', 'On')			
Command	Value	Value	Value
Input	'Video / SCART 1'	'HDMI 1'	'HDMI 2'
	'HDMI 3'	'HDMI 4'	'PC'
# Input example InterfaceName.Set('Input', 'Video / SCART 1')			
Command	Value	Value	Value
MenuNavigation	'Up'	'Down'	'Left'
	'Right'	'Select'	'Return'
	'Home'		
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Up')			
Command	Value	Value	
Power	'On'	'Off'	
# Power example InterfaceName.Set('Power', 'On')			
Command	Value	Value	
VideoMute	'On'	'Off'	
# VideoMute example InterfaceName.Set('VideoMute', 'On')			
Command	Value		
Volume	0 to 100 in steps of 1		
# Volume example InterfaceName.Set('Volume', 100)			

Status Available (Ethernet)

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},
```

FeedbackHandler)

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command	Value	Value
AudioMute	'On'	'Off'
# AudioMute example InterfaceName.Update('AudioMute') Value = InterfaceName.ReadStatus('AudioMute') InterfaceName.SubscribeStatus('AudioMute', None, FeedbackHandler)		
Command	Value	Value
ConnectionStatus	'Connected'	'Disconnected'
# ConnectionStatus example Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)		
Command	Value	Value
Input	'Video / SCART 1'	'HDMI 1'
	'HDMI 3'	'HDMI 4'
		'HDMI 2'
		'PC'
# Input example InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)		
Command	Value	Value
Power	'On'	'Off'
# Power example InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)		
Command	Value	Value
VideoMute	'On'	'Off'
# VideoMute example InterfaceName.Update('VideoMute') Value = InterfaceName.ReadStatus('VideoMute') InterfaceName.SubscribeStatus('VideoMute', None, FeedbackHandler)		
Command	Value	
Volume	0 to 100 in steps of 1	
# Volume example InterfaceName.Update('Volume') Value = InterfaceName.ReadStatus('Volume') InterfaceName.SubscribeStatus('Volume', None, FeedbackHandler)		

Cable and Adapter Requirements

Captive Screw to 3.5mm TRS RS-232 Serial Cable.

Notes for the Device

To Enable remote device control over RS232, make sure that the **RS232C control** is set as follows:

- [HOME] Settings → RS232C control → Via serial port

You must then switch to **Normal mode** or **Pro mode** because this function does not work in **Pro settings mode** when **Via serial port** is set to a setting other than **Via HDMI port**.

Serial communication

Port Type: RS-232

Baud Rate: 9600

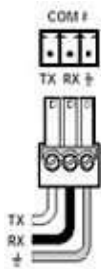
Data Bits: 8

Parity: None

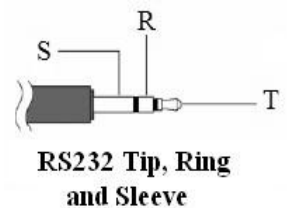
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TXD		T	TXD
RXD		R	RXD
GND		S	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scriptor ethernet interface.

Port Type: Ethernet (TCP)

Default Port: 20060

Logon Credentials Supported: No

Multi-Connection Capabilities: Undetermined

Port Changeability: No

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

To Enable remote device control over Ethernet, make sure that the **IP control** & the **Control remotely** are set as follows:

- [HOME] Settings → Network and Internet → Remote device settings → Control remotely → Enable [Default: Enabled]
- [HOME] Settings → Network and Internet → Home network setup → IP control → Simple IP control → Enable [Default: Disabled]

Appendix A. Set Commands(Serial)

Aspect Ratio 16:9	\x8C\x00F\x03\x01\x00\xD6
Aspect Ratio 4:3	\x8C\x00F\x03\x01\x04\xDA
Aspect Ratio 4:3 Wide Zoom	\x8C\x00F\x03\x01\x03\xD9
Audio Mute Off	\x8C\x006\x03\x01\x00\xC6
Audio Mute On	\x8C\x006\x03\x01\x01\xC7
AV Mute Off	\x8C\x00\x06\x03\x01\x00\x96
AV Mute On	\x8C\x00\x06\x03\x01\x01\x97
Input HDMI 1	\x8C\x00\x02\x03\x04\x01\x96
Input HDMI 2	\x8C\x00\x02\x03\x04\x02\x97
Input HDMI 3	\x8C\x00\x02\x03\x04\x03\x98
Input HDMI 4	\x8C\x00\x02\x03\x04\x04\x99
Input PC	\x8C\x00\x02\x03\x05\x01\x97
Input Shared Input	\x8C\x00\x02\x03\x07\x01\x99
Input Video / SCART 1	\x8C\x00\x02\x03\x02\x01\x94
Menu Navigation Down	\x8C\x00g\x03\x01u1
Menu Navigation Home	\x8C\x00g\x03\x01`W
Menu Navigation Left	\x8C\x00g\x03\x014+
Menu Navigation Return	\x8C\x00g\x03\x97#\xB0
Menu Navigation Right	\x8C\x00g\x03\x013*
Menu Navigation Select	\x8C\x00g\x03\x01e\
Menu Navigation Up	\x8C\x00g\x03\x01tk
Power Off	\x8C\x00\x00\x02\x00\x8E
Power On	\x8C\x00\x00\x02\x01\x8F
Video Mute Off	\x8C\x00\x0D\x03\x01\x00\x9D
Video Mute On	\x8C\x00\x0D\x03\x01\x01\x9E
Volume 0	\x8C\x00\x05\x03\x01\x00\x95
Volume 100	\x8C\x00\x05\x03\x01d\xF9

Appendix B. Update Commands(Serial)

AV Mute	\x83\x00\x06\xff\xff\x87
Input	\x83\x00\x02\xff\xff\x83
Power	\x83\x00\x00\xff\xff\x81
Volume	\x83\x00\x05\xff\xff\x86

Appendix C. Set Commands(Ethernet)

Audio Mute Off	*SCAMUT0000000000000000\x0A
Audio Mute On	*SCAMUT0000000000000001\x0A
Input HDMI 1	*SCINPT00000000100000001\x0A
Input HDMI 2	*SCINPT00000000100000002\x0A
Input HDMI 3	*SCINPT00000000100000003\x0A
Input HDMI 4	*SCINPT00000000100000004\x0A
Input PC	*SCINPT00000000500000001\x0A
Input Video / SCART 1	*SCINPT00000000300000001\x0A
Menu Navigation Down	*SCIRCC0000000000000010\x0A
Menu Navigation Home	*SCIRCC0000000000000006\x0A
Menu Navigation Left	*SCIRCC0000000000000012\x0A
Menu Navigation Return	*SCIRCC0000000000000008\x0A
Menu Navigation Right	*SCIRCC0000000000000011\x0A
Menu Navigation Select	*SCIRCC0000000000000013\x0A
Menu Navigation Up	*SCIRCC0000000000000009\x0A
Power Off	*SCPOWR0000000000000000\x0A
Power On	*SCPOWR0000000000000001\x0A
Video Mute Off	*SCPMUT0000000000000000\x0A
Video Mute On	*SCPMUT0000000000000001\x0A
Volume 0	*SCVOLU0000000000000000\x0A
Volume 100	*SCVOLU00000000000000100\x0A

Appendix D. Update Commands(Ethernet)

Audio Mute	*SEAMUT#####\x0A
Input	*SEINPT#####\x0A
Power	*SEPOWR#####\x0A
Video Mute	*SEPMUT#####\x0A
Volume	*SEVOLU#####\x0A